

PLC-Controlled Electromagnetic Pick-and-Place System

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Abstract—This report details the mechanical and electrical hardware integration for a Programmable Logic Controller (PLC) based electromagnetic pick-and-place system. The technical focus was directed toward electrical wiring, power distribution, and the establishment of PLC I/O connections to ensure a robust interface between control logic and physical actuators. The system architecture was developed using an LS Electric XBC-DN20SU PLC to coordinate three-axis motion via stepper drivers and a linear actuator. Key hardware deliverables included the design of a safety-compliant industrial enclosure following UL 508A standards and the implementation of dual-rail 24 V DC power distribution to isolate inductive noise from control logic. Through the integration of custom 3D-printed PETG mounts and hardware-level emergency stop loops, the prototype was transitioned from a hobbyist frame to a functional industrial automation system. Validation testing confirmed that the hardware integration supported repeatable pick-and-place cycles with a motion resolution of 0.012 mm/step and maintained load retention during safety interrupts.

Index Terms—Human-machine interface (HMI), programmable logic controller (PLC), XG5000, XP-Builder, limit switch, actuator, electromagnet, pick-and-place system.

I. INTRODUCTION

Precision motion control serves as the cornerstone of modern industrial automation, facilitating applications from CNC machining to automated logistics. This project focuses on the design and implementation of a three-axis pick-and-place prototype utilizing an LS Electric XBC-DN20SU PLC. Theoretical foundations for the system are rooted in industrial standards for motion control and electrical safety, where the controller must interface seamlessly with power electronics and mechanical actuators. The physical realization of the system ensures that the electrical architecture supports high-speed pulse outputs while maintaining rigorous safety standards, such as voltage segregation and hardware-level interlocking.

II. PROJECT GOALS

The primary objective was the creation of a working automation system capable of manual jog control and automatic motion sequences. Specific technical goals included:

- Develop custom 3D-printed brackets for end-effector and motion-limit sensor integration.
- Design the system as two discrete subsystems: the mechanical gantry and the control panel.
- Design a control enclosure following UL 508A standards and NEC Article 409 to maintain physical separation between power and signal lines.

- Implement a hardware-level E-Stop loop to ensure immediate motion cessation without dropping the payload.

III. MECHANICAL AND ELECTRICAL HARDWARE INTEGRATION

A. Physical System Modification and Custom Integration

The prototype was developed by modifying a SparkFun Shapeoko gantry frame, utilizing the motion actuators (X -axis stepper motor and dual Y -axis stepper motors) and chassis. To achieve the desired operation, custom mounts were engineered and 3D printed using high-density PETG (see Fig. 1):

- **Actuator and End-Effector Mounts:** Custom housings secure the linear actuator to the X -axis and couple the electromagnet to the Z -axis.
- **Limit Switch Brackets:** Specialized brackets mount mechanical micro-switches for homing and overtravel protection.
- **Gantry Junction Box:** A local termination point for all motors and actuators, minimizing wire count crossing the motion axes.

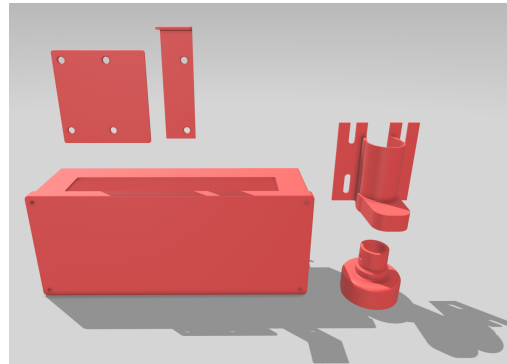


Fig. 1. 3D printed components including limit switch mounts, secondary enclosure, and actuator mounts.

B. Electrical Design and Schematic Documentation

The system's electrical architecture was designed using SkyCAD Electrical to ensure industrial-grade documentation. As shown in Fig. 2, the schematic is organized into a comprehensive set:

- **Power and Safety:** Details the dual-rail 24 V DC strategy and the hardwired E-Stop loop.

- **Motion Control:** Illustrates the dual-motor Y -axis and single-motor X -axis wiring.
- **Load Isolation:** Documents the relay isolation used to protect the PLC.
- **PLC I/O Mapping:** Defines the sinking I/O structure and limit switch configuration.

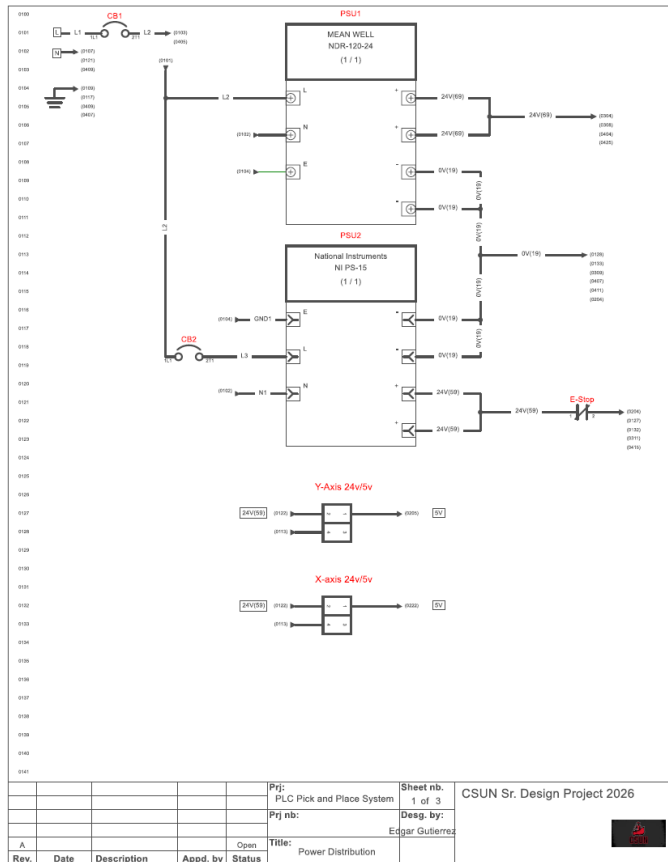


Fig. 2. Primary electrical schematic detailing power distribution and the safety loop logic.

C. Industrial Enclosure and Electrical Architecture

The electrical system is housed in an impact-resistant enclosure (see Fig. 3) organized according to NEC Article 409 requirements for industrial control panels:

- **Control and Interface Logic:** Houses the LS Electric XBC-DN20SU PLC and the XP40TTA/DC HMI units.
- **Power Distribution and Protection:** Managed through dual 24 V, 5 A power supplies and primary circuit breakers. These breakers were selected according to IEEE 1015 guidelines to ensure reliable overcurrent protection and facilitate modular maintenance.
- **Motion Drive and Isolation Subsystem:** High-resolution TB6600 stepper drivers interface between the PLC and actuators. An interposing relay board provides electrical isolation following IEEE C37.90 principles to protect the control logic from inductive apparatus interference.
- **Termination and Wire Management:** Wiring is consolidated through labeled landing terminals using industrial

color-coding: blue for 24 V signals, red for outputs, white for 0 V common, green for ground, and black/white for 120 V AC.

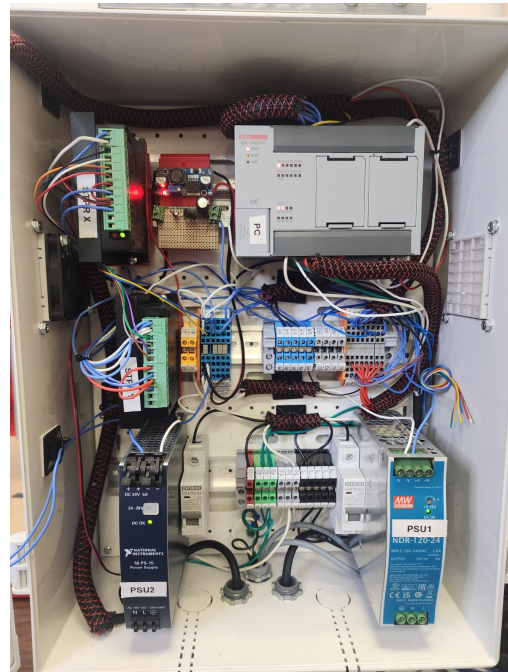


Fig. 3. Control panel layout: PLC, stepper drivers, power supplies, and segregated terminals.

D. Power Supply Selection and Demand Analysis

The power architecture was designed to accommodate the peak simultaneous demand of all axes while maintaining logic stability. To minimize system complexity, all electromechanical components were specifically selected to operate on a unified 24 V DC standard. A dual-power-supply strategy was implemented to isolate high-current motion noise from sensitive control logic. This configuration involved a strategic distribution of loads: PSU 1 was dedicated to powering the limit switches and the electromagnet, while PSU 2 supported the stepper motors and the Z -axis linear actuator. Each supply was integrated with a separate circuit breaker to facilitate maintenance and troubleshooting. This modular design allowed for the isolation of specific subsystems, enabling hardware modifications or diagnostic testing without requiring a complete system power-down.

As detailed in Table I, the dual Y -axis motors were connected to a single stepper driver configured for 1.5 A. In this parallel arrangement, the current was distributed between the motors, providing approximately 0.75 A each. Testing confirmed that this current level provided sufficient torque for the intended work envelope without necessitating driver reconfiguration. Furthermore, the Z -axis linear actuator—originally designed for high-force applications such as powered furniture—operates well below its 3.0 A rated capacity in this system. Because the compact work envelope restricts the lifting load to lightweight workpieces, the actual current draw remains minimal, ensuring the motion bus remains stable

within the 5.0 A supply threshold during all phases of operation.

TABLE I
SYSTEM POWER BUDGET AND COMPONENT DEMAND.

Component	Voltage	Peak Current	Power
Stepper Motors (X, Y)	24 V DC	3.0 A (1.5 A ea.)	72 W
Linear Actuator (Z)	24 V DC	3.0 A (stall)	72 W
Electromagnet	24 V DC	0.25 A	6 W
PLC, HMI, and Relays	24 V DC	0.50 A	12 W
Total Peak	24 V DC	6.75 A	162 W

IV. EVALUATION OF ENGINEERING DESIGN SOLUTIONS

Engineering solutions were evaluated based on safety, welfare, and economic feasibility. The selection of a 24 V DC linear actuator and electromagnet was an economical decision, providing a functional alternative to expensive pneumatic systems or additional motion modules. Additionally, utilizing high-density PETG for custom mounts allowed for rapid functional prototyping and reduced material waste while maintaining the structural integrity of the gantry.

To ensure hardware longevity, a secondary relay board (see Fig. 4) was integrated to create a physical air gap between the PLC outputs and inductive flyback from the actuator and magnet. Furthermore, gantry-mounted terminal blocks were implemented to localize wiring, minimizing the number of conductors crossing the motion axes and reducing the risk of mechanical fatigue or wire snagging. Regarding safety logic, a critical design choice was made to power the electromagnet through a dedicated source independent of the motion-safety rail. This ensures the payload remains secure during an Emergency Stop event, preventing "dropped load" hazards and adhering to industrial safety standards.

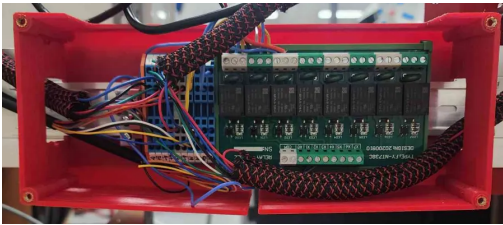


Fig. 4. Gantry mounted relay board and landing terminals.

V. IMPACT OF ENGINEERING SOLUTIONS IN GLOBAL CONTEXTS

The decision to utilize standardized industrial hardware—specifically the LS Electric PLC and DIN rail architecture—ensures that this system is compatible with international standards like IEC 60204-1. This design choice allows the control logic and wiring footprint to be deployed in various global economic contexts with minimal reconfiguration, supporting a shift toward modular production facilities.

Furthermore, the implementation of a hardware-level safety loop addresses critical industrial welfare concerns. By removing human operators from the active motion envelope and replacing manual tasks with a mechatronic system, the

risk of human error and contamination is mitigated. This is particularly vital in high-precision sectors such as microchip placement, where even minor human intervention can lead to significant yield loss. Additionally, the architecture's inherent scalability allows these solutions to be adapted for large-scale logistics or hazardous waste management. By prioritizing reliable, hardware-interlocked safety over software-only solutions, this design establishes a framework for safer, more efficient manufacturing environments worldwide.

VI. COLLABORATIVE ENVIRONMENT AND PROJECT LEADERSHIP

A cohesive and inclusive working environment was established by prioritizing technical transparency and mutual support throughout the design lifecycle. Leadership contributions focused on providing the technical infrastructure necessary for the group's success, which included sourcing all compatible hardware components and validating their integration through bench testing and wiring. To facilitate a collaborative programming workflow, sample logic routines were developed and shared, serving as a baseline for the team's manual and automatic sequence programming.

Project goals and benchmarks were established by defining the system's operational logic and safety requirements during the initial planning phase. A specific emphasis was placed on industrial welfare through the integration of a hardware-level Emergency Stop, ensuring a safe testing environment for all members. Throughout the semester, regular communication was maintained to ensure technical synchronization across subsystems. By offering technical support to teammates during the integration of the PLC and HMI logic and providing comprehensive I/O documentation, the group's collective efforts were streamlined, resulting in the successful meeting of all project objectives.

VII. TESTING RESULTS AND VALIDATION

A. Hardware Refinement and Signal Logic

A primary technical challenge identified during validation involved the integration of the PLC's sinking output architecture with the interposing relay board. The LS Electric PLC utilizes NPN outputs that complete the circuit to 0 V. To accommodate this, the relay board power distribution was re-engineered to utilize a common +24 V DC rail. By rewiring the output terminals to deliver 0 V to the coils, the relays successfully energized, maintaining the required electrical isolation.

B. Functional Performance

Testing confirmed a motion resolution of 0.012 mm/step. The HMI command mapping and limit switch protection passed all functional test cases, verifying the hardware provides a reliable platform for autonomous cycles.

VIII. CONCLUSION

The integration of industrial hardware with motion control logic successfully met all senior design requirements. The development of the mechanical and electrical infrastructure provided a stable platform for the PLC and HMI subsystems to function. Future improvements may include pneumatic actuator integration for high-speed applications and Ethernet-based communication for remote monitoring.

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